

2.3 Environmental threats to our Planet

Climate change and extreme weather conditions cause many threats to both people and the environment. This theme develops understanding of these key environmental threats affecting countries and the world as a whole. Learners will explore the changing climate, including possible causes, and the

current consequences. An introduction to the global circulation of the atmosphere leads to a study of extreme weather conditions and subsequent drought which can impact both people and the environment at a range of scales.

Section	Key Idea	Content	Scale
2.3.1	The climate has changed from the start of the Quaternary period.	- Overview of how the climate has changed from the beginning of the Quaternary period to the present day, including ice ages. - Key periods of warming and cooling since 1000AD, including the medieval warming, Little Ice Age and modern warming. - Evidence for climate change over different time periods, including global temperature data, ice cores, tree rings, paintings and diaries.	G
2.3.2	There are a number of possible causes of climate change.	- Theories of natural causes of climate change including variations in energy from the sun, changes in the Earth's orbit and volcanic activity. - How human activity is responsible for the enhanced greenhouse effect which contributes to global warming.	G
2.3.3	Climate change has consequences.	Summary of a range of consequences of climate change currently being experienced across the planet.	G, R, N, L
2.3.4	The global circulation of the atmosphere controls weather and climate.	- Distribution of the main climatic regions of the world. - Outline how the global circulation of the atmosphere is controlled by the movement of air between the poles and the equator. - How the global circulation of the atmosphere leads to extreme weather conditions (wind, temperature, precipitation) in different parts of the world.	G, R
2.3.5	Extreme weather conditions cause different natural weather hazards.	- Outline the causes of the extreme weather conditions that are associated with the hazards of tropical storms and drought. - The distribution and frequency of tropical storms and drought, and whether these have changed over time.	G
2.3.6	Drought can be devastating for people and the environment.	Case study of one drought event caused by El Niño/La Niña: - how the extreme weather conditions of El Niño/La Niña develop and can lead to drought - effects of the drought event on people and the environment - ways in which people have adapted to drought in the case study area.	G, R, N, L